Technical Note – Novel Condensates with High Bio-Based Content as Syntan Replacements

PDF (https://journals.uc.edu/index.php/JALCA/article/view/8813/7013)

Published: 2025-07-01

DOI: https://doi.org/10.34314/kc29s514 (https://doi.org/10.34314/kc29s514)

Jochen Ammenn

Abstract

Syntans are widely used in leather making. They are made from fossil-based phenol and formaldehyde. Both building blocks are toxic. Many syntans currently have high values of bisphenols, some have a problem with rest-monomeric formaldehyde. This is why the search is ongoing for re-tanning agents made from less toxic and preferably renewable starting materials, that can substitute for syntans avoiding formaldehyde and bisphenol issues.

In this article novel condensates are introduced that are obtained by condensation of dicarboxylic esters with renewable aromatic aldehydes. A subsequent reaction leads to complete water solubility. A preferred aromatic aldehyde is vanillin, which is commonly used in nutrition, avoiding any toxicity issue. Vanillin can be obtained from a waste stream of the paper industry.

In application on leather, these novel condensates could achieve or outperform the performance of syntans and vegetable tanning agents in re-tanning concerning softness, fullness, and fastness properties. No formaldehyde was used in the condensation, no bisphenols formed. The synthesis is facile and dominantly renewable starting materials were chosen as building blocks. These results have been presented at the 12th Ledertage conference in Salzburg, Austria in June 2024.

Additionally, this article will present for the first time, results of the application of these condensates in sole tanning

Issue

Vol. 120 No. 7 (2025): Journal of American Leather Chemists Association (https://journals.uc.edu/index.php/JALCA/issue/view/619)

Section

Articles

Information

For Readers (https://journals.uc.edu/index.php/JALCA/information/readers)

For Authors (https://journals.uc.edu/index.php/JALCA/information/authors)

For Librarians (https://journals.uc.edu/index.php/JALCA/information/librarians)

Subscription

Login to access subscriber-only resources.

Platform & workflow by OJS / PKP (https://journals.uc.edu/index.php/JALCA/about/aboutThisPublishingSystem)